Glaucoma Detection by Combining SLO and OCT Fundus Images: An Ensemble of Pretrained Two-Branch EfficientNets

Rushikesh

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1 Introduction

Task. Glaucoma is a chronic optic-nerve disease and a leading cause of irreversible blindness, so automated screening from retinal imaging is valuable. We address one problem from the FairDomain benchmark [1]: *given two grayscale images of the same eye, an SLO fundus image and an OCT-derived fundus image, predict whether the eye has glaucoma* (binary classification). A solution outputs a probability $P(\text{glaucoma})$ for each image pair. It is judged by the area under the ROC curve (AUC) on a fixed held-out test set and, because the data carry demographic attributes, by how evenly that performance is spread over gender, race and ethnicity groups (Section 4.3).

1.1 The data

Contents. The release holds 5,000 samples in a fixed split of 3,000 training and 2,000 test samples. Each sample is one `.npz` file with the two images (`slo_fundus` and `oct_fundus`; each 224×224, 8-bit, single channel), the label `glaucoma` (0 or 1), and patient attributes: age, gender, race, ethnicity, language, marital status and the visual-field mean deviation (MD). A CSV file indexes the split and the attributes. No segmentation masks are included, so the task is classification only.

What the images look like. Figure 1 shows random samples of each class. Both modalities show the optic disc and the retinal vessels around it. The SLO fundus image is darker, with uneven illumination and dark patches; the OCT fundus image is brighter and smoother. The two modalities therefore differ clearly in appearance, but *the two classes do not look different to the eye*, and simple intensity statistics do not separate them either: over 300 random training images the mean pixel value is 73 (no glaucoma) against 70 (glaucoma) for SLO, and 169 against 169 for OCT. The signal is subtle structural information that a model has to learn.

Labels and demographics. Glaucoma is the positive class in 44% of both splits (1,317 of 3,000 training and 887 of 2,000 test samples), a mild imbalance that is one reason we report AUC rather than accuracy. The demographic groups are strongly imbalanced (Table 1): 73% of samples are female, 77% white, 16% Black, 8% Asian and 4% Hispanic. Only 83 test images belong to Hispanic patients, so any estimate for that group is very noisy.

Clinical variables. MD summarises a visual-field test (in dB; 0 is normal, more negative means more field loss). It averages 0.1 ± 0.8 in non-glaucoma and -8.2 ± 5.7 in glaucoma training samples, so it almost encodes the label: giving it to the model raises test AUC to 0.958 (Appendix F). We treat it as a leakage risk and use it only in one clearly labelled upper-bound run, never in a reported

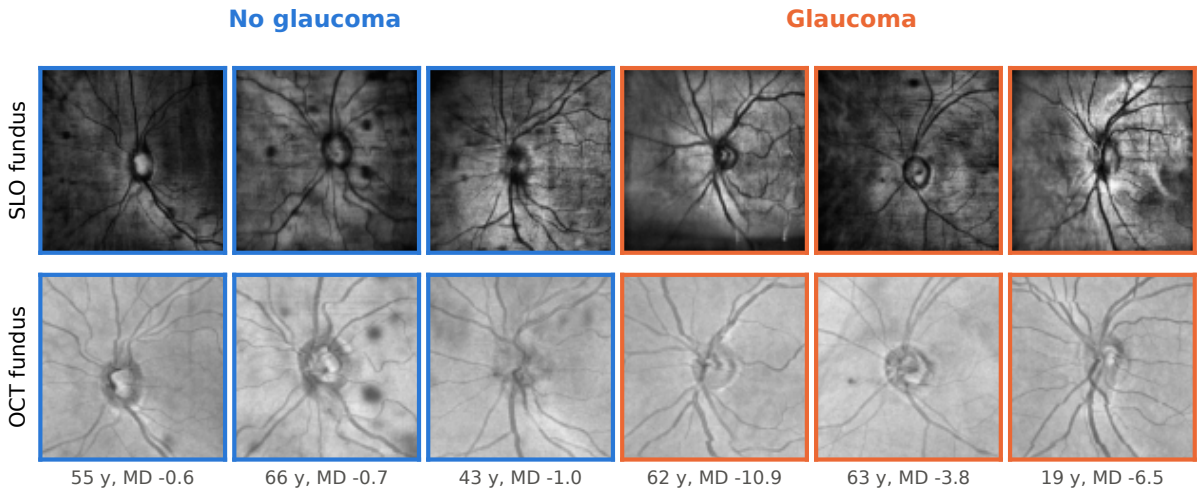


Figure 1: Randomly chosen training samples (seed 0, not hand-picked) for each class. Top row: SLO fundus; bottom row: OCT fundus. Under each pair: the patient's age and visual-field mean deviation (MD, dB). Blue frame: no glaucoma; orange frame: glaucoma.

Split	N	Glaucoma	Female	White	Black	Asian	Hispanic
Training	3000	1317 (44%)	2204 (73%)	2308 (77%)	467 (16%)	225 (8%)	121 (4%)
Testing	2000	887 (44%)	1449 (72%)	1522 (76%)	301 (15%)	177 (9%)	83 (4%)

Table 1: Dataset composition. Percentages are of each split.

comparison. Age differs by class (57.8 against 64.4 years on average) but adding it did not help (Appendix F).

What an engineer should keep in mind.

- **Small data:** after holding out validation data only 2,550 training images remain, so overfitting is the main difficulty (Appendix C).
- **Small test set:** 2,000 images give AUC confidence intervals of about ±0.02; differences smaller than that between two models are not meaningful.
- **No official validation set:** we hold out a stratified 15% of the training images for all model and configuration selection (Section 4.1).
- **Two different views:** a model trained on one modality transfers poorly to the other (Appendix F), so the modalities must be handled separately and combined.
- **Fairness-aware evaluation:** results should be reported per demographic group, not only overall.

Our approach and result. Our final model is an *ensemble of eight ImageNet-pretrained two-branch EfficientNet-B0 networks*, one branch per modality, trained with five-fold cross-validation plus three runs on all training data, with horizontal-flip test-time averaging. It reaches a test AUC of 0.834 (95% CI 0.816–0.852) against 0.717 for a from-scratch ResNet18 baseline. The main

text describes only this model and the experiments that justify it; the other experiments (a 13-model backbone comparison, a training-recipe study with ablations, fusion variants, a foundation model, fairness losses, cross-domain transfer) are documented with their motivation and setup in the appendices.

Contributions. (1) A simple, strong pipeline for SLO+OCT glaucoma classification with bootstrap confidence intervals and per-group results. (2) A controlled demonstration, on the same backbone and recipe, that using both modalities beats either alone (0.827 against 0.758 and 0.764). (3) Evidence that averaging models trained on different data splits gains about 0.03 AUC over its members. (4) An honest account, in the appendices, of what did *not* help, including a hypothesis of ours that the ablations refuted.

2 Related Work

Fairness-oriented ophthalmic datasets. The Harvard Ophthalmology AI Lab released a series of datasets pairing retinal imaging with demographic attributes. Harvard Glaucoma Fairness [2] introduced a glaucoma dataset with 2D and 3D imaging and proposed Fair Identity Normalization (FIN), which equalises feature importance across identity groups. Harvard Eye Fairness [3] extended this to 30,000 subjects with 3D OCT and proposed Fair Identity Scaling (FIS), a loss weighting that combines individual and group scaling. FairCLIP [4] studies fairness in vision–language learning on related data. FairDomain [1], whose data we use, studies fairness under *domain shift* for both segmentation and classification, treating different image types as domains, and proposes a fair identity attention (FIA) module. Group-wise AUC, worst-group AUC and equity-scaled AUC (ES-AUC) are the fairness-aware metrics commonly reported in this line of work.

Recent glaucoma-specific work. FairEnc [5] builds a fair vision–language model on SLO images paired with clinical notes, Fair-Eye Net [6] combines fundus photographs, OCT-derived metrics and visual-field indices with a fairness constraint, and RED-Sphere [7] targets cross-population domain generalisation for fundus disease. We did not reproduce these methods; they motivate our per-group evaluation.

Backbones and training techniques. Our final model uses EfficientNet [9] backbones with ImageNet [19] pretraining from the `timm` library [20]. We also evaluated residual networks [8], DenseNet [10], RegNet [11], ConvNeXt [12], ViT [13], DeiT [14], Swin [15], CoAtNet [16], MaxViT [17] and DINOv2 [18] (Appendix B). The regularisers and training tools involved are AdamW [22] with a one-cycle schedule [23], label smoothing [27], stochastic depth (drop-path) [26], and, in the appendices, mixup [24], CutMix [25] and gradient reversal [28].

3 Method

3.1 Problem formulation

Each sample is a pair $(\mathbf{x}_S, \mathbf{x}_O)$ of 224×224 single-channel images (SLO fundus and OCT fundus), a binary label $y \in \{0, 1\}$ (glaucoma), and demographic attributes g used only for evaluation. We learn $p(y = 1 \mid \mathbf{x}_S, \mathbf{x}_O)$.

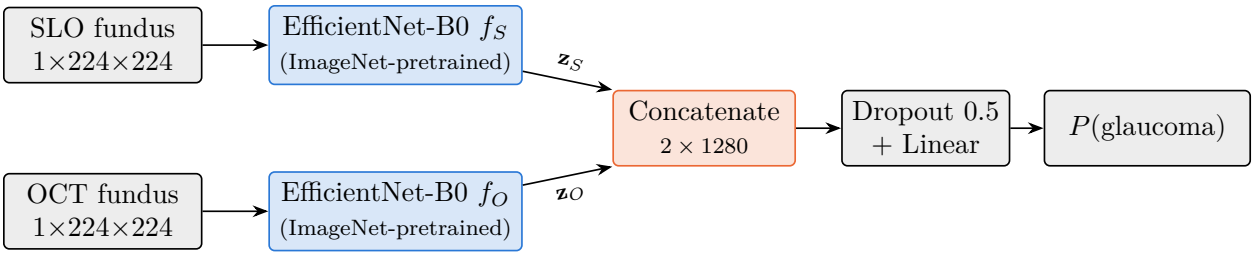


Figure 2: One ensemble member: a two-branch late-fusion network with an EfficientNet-B0 backbone per modality.

3.2 Two-branch EfficientNet-B0

Figure 2 shows the network. Each modality is processed by its own EfficientNet-B0 backbone (f_S, f_O; separate weights, initialised from ImageNet-1k, 4.0 M parameters each). The pooled 1280-dimensional features are concatenated, passed through dropout (rate 0.5) and a linear layer, giving two logits. Pretrained first-layer filters are adapted to one input channel by `timm`. Pixels are scaled to $[0, 1]$ and normalised as $(x - 0.5)/0.25$. Appendix D describes the alternative fusion designs we tried (early, gated, cross-attention); none outperformed plain concatenation clearly enough to use.

3.3 Training recipe

Members are trained with cross-entropy with label smoothing 0.1 and AdamW [22] with a one-cycle schedule [23] (peak learning rate 5×10^{-4}, 5× larger for the new head, weight decay 10^{-2}, batch size 32, 25 epochs, gradient clipping at 1.0). Regularisation consists of drop-path [26] with rate 0.2, head dropout 0.5, an exponential moving average (EMA) of the weights (decay 0.995) used for evaluation, and on-GPU augmentation: horizontal flips, rotation (±15°), scaling (±10%), translation (up to 5% of the image size) and brightness/contrast jitter. This configuration was selected automatically from twelve runs of the recipe study by *validation* AUC (Appendix C): it beat the plain recipe's validation AUC (0.852 against 0.832 averaged over three seeds) by more than the seed-to-seed spread.

3.4 Ensemble and test-time augmentation

The final predictor averages the probabilities of eight members (Figure 3):

- **Five cross-validation members.** The 3,000 training images are split into five stratified folds. Member k trains on four folds (2,400 images) and uses the held-out fold (600 images) to choose its best epoch.
- **Three full-data members.** Three seeds trained on all 3,000 images, with no validation set; the final-epoch weights are used.

At test time each member's prediction is averaged with its prediction on the horizontally flipped image (test-time augmentation, TTA), and the member probabilities are averaged. The ensemble composition was fixed before looking at test results; it is one of four ensembles we evaluated (Appendix G).

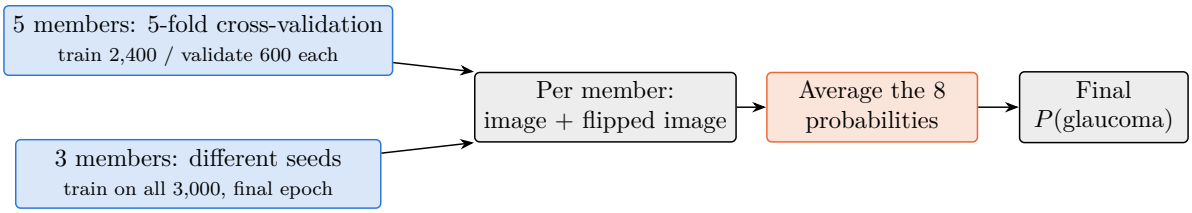


Figure 3: The final ensemble: eight EfficientNet-B0 two-branch models, each evaluated on an image and its mirror image, with probabilities averaged.

Hyper-parameter	Value
Backbone	EfficientNet-B0 $\times 2$ (ImageNet-1k, `ra_in1k` weights)
Optimiser / schedule	AdamW, one-cycle (15% warm-up), gradient clip 1.0
Learning rate (body / head)	5×10^{-4} / 2.5×10^{-3}
Weight decay, batch size, epochs	10^{-2}, 32, 25
Label smoothing, drop-path, head dropout	0.1, 0.2, 0.5
EMA decay	0.995
Augmentation	flip, rotation $\pm 15°$, scale $\pm 10\%$, shift $\leq 5\%$, brightness/contrast jitter
Test-time augmentation	horizontal flip

Table 2: Hyper-parameters of every ensemble member.

4 Experiments

4.1 Data splits

The data are described in Section 1.1 (Table 1). From the 3,000 training images we hold out a stratified 15% (450 images) as a *validation* set, used for model and configuration selection; the remaining 2,550 images train the single-run models. The five cross-validation members instead use five stratified folds of the 3,000 images, and the three full-data members use all of them. The 2,000 test images are never used for training or for automatic selection.

4.2 Implementation and hyper-parameters

All models are implemented in PyTorch [21] with `timm` [20] and trained in fp32 on a single Tesla P100 (16 GB); one member trains in about 11–12 minutes. The hyper-parameters are those of Section 3.3 and are summarised in Table 2.

4.3 Evaluation protocol

The main metric is the area under the ROC curve (AUC) on the 2,000 test images, which is threshold-free and insensitive to the 44/56 class balance. To judge uncertainty we report non-parametric bootstrap 95% confidence intervals over test images (1,000 resamples). Intervals are about ± 0.02 wide, so differences smaller than that between two models are not meaningful. For fairness we report AUC within each gender, race and ethnicity group, the *worst-group AUC* (minimum over gender and race groups with at least 100 test samples), and the equity-scaled AUC (ES-AUC)

$$\text{ES-AUC} = \frac{\text{AUC}}{1 + \sum_g |\text{AUC} - \text{AUC}_g|}, \tag{1}$$

Model	Input	Test AUC	95% CI	+ flip TTA	ES-AUC race	Worst-group AUC
ResNet18, two-branch, from scratch (baseline)	SLO+OCT	0.717	[0.694, 0.739]	0.725	0.696	0.706
EfficientNet-B0, SLO only	SLO	0.758	[0.736, 0.778]	0.762	0.727	0.723
EfficientNet-B0, OCT only	OCT	0.764	[0.742, 0.784]	0.771	0.750	0.753
EfficientNet-B0, two-branch, plain recipe (one run)	SLO+OCT	0.827	[0.808, 0.844]	0.828	0.815	0.816
EfficientNet-B0, two-branch, final recipe (one run)	SLO+OCT	0.817	[0.797, 0.835]	0.820	0.793	0.803
five cross-validation members, mean ± s.d.	SLO+OCT	0.800 ± 0.008	–	–	–	–
Ensemble of the 5 cross-validation models	SLO+OCT	0.830	[0.811, 0.848]	0.831	0.807	0.820
Final: ensemble of 8 models	SLO+OCT	0.834	[0.816, 0.852]	0.835	0.802	0.821

Table 3: Main results on the 2,000 test images. "Plain recipe" = flips and mild jitter, 15 epochs; "final recipe" = Section 3.3. TTA = horizontal-flip test-time averaging.

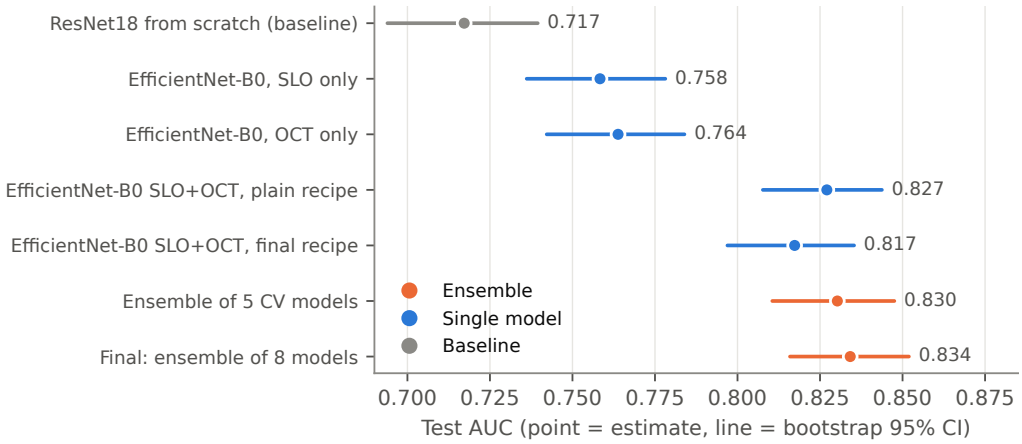


Figure 4: Test AUC with bootstrap 95% confidence intervals for the baseline, single models and ensembles.

computed separately per attribute (our implementation of the equity-scaled AUC used in the Harvard fairness benchmarks; absolute values may differ from the original definition).

4.4 Baselines

- **ResNet18 from scratch:** the same two-branch late-fusion design with random initialisation (plain recipe, 15 epochs).
- **Single-modality EfficientNet-B0:** the same backbone and plain recipe on SLO only and on OCT only.
- **Single two-branch EfficientNet-B0:** one run with the plain recipe (from our first backbone comparison) and one run with the final recipe.
- **Cross-validation members:** the five individual members, to measure the gain from ensembling on identical models.

4.5 Results

Table 3 and Figure 4 give the main results.

Both modalities matter. With the same EfficientNet-B0 backbone and recipe, SLO alone reaches 0.758 and OCT alone 0.764, whereas fusing both reaches 0.827, a gain of about 0.06–

n (test)	Female 1449	Male 551	Asian 177	Black 301	White 1522	Non-Hisp. 1917	Hispanic 83
ResNet18 from scratch (baseline)	0.721	0.706	0.711	0.735	0.710	0.723	0.556
EfficientNet-B0, final recipe (one run)	0.816	0.822	0.831	0.803	0.814	0.819	0.811
Final: ensemble of 8 models	0.832	0.841	0.856	0.821	0.829	0.836	0.806

Table 4: Test AUC within demographic groups.

0.07 that is well outside the confidence intervals. The two images carry different information (Appendix F shows that a model trained on one modality transfers poorly to the other).

Pretrained two-branch EfficientNets versus the baseline. The from-scratch ResNet18 baseline reaches 0.717, whereas the pretrained two-branch EfficientNet-B0 reaches 0.817–0.827 in single runs. This difference mixes pretraining and architecture; Appendix B shows that every pretrained backbone we tried scored higher than the scratch baseline, but we did not train scratch versions of the other backbones, so we cannot attribute the gap to pretraining alone.

Ensembling. The five cross-validation members score 0.800 ± 0.008 individually (range 0.789–0.808). Averaging them gives 0.830, a gain of 0.030 on models that are otherwise identical. Adding the three full-data members gives the final 0.834 [0.816, 0.852]; flip TTA adds a further 0.001. The ensemble is not clearly better than the best single run (0.827, plain recipe, one seed): that difference lies well inside the confidence intervals. The practical advantage is stability, because an ensemble does not depend on one fortunate run (Appendix C indicates that the 0.827 single run is probably favourable luck).

Fairness across groups. Table 4 shows AUC within each group. For the final ensemble, AUC is 0.82–0.86 across gender and race groups, with a worst-group AUC of 0.821, against 0.706 for the baseline. The ES-AUC for race is 0.802. The Hispanic group (83 test images) has the lowest AUC (0.806 for the ensemble) but its confidence interval is very wide, so we make no claim about it. Fairness-specific training losses did not help beyond this (Appendix E).

4.6 Limitations

(i) The test set has 2,000 images, so confidence intervals are about ± 0.02; the ensemble's lead over the best single model is within them. (ii) We evaluated four ensemble compositions and report the one with the highest test AUC as the final model, so its figure is mildly optimistic (the four ranged from 0.816 to 0.834; Appendix G). (iii) Automatic selection used validation data only, but we read test results while planning later experiments. (iv) The ablations in Appendix C found that replacing the geometric augmentation by flips alone raises single-model AUC by about 0.02; the final ensemble does not use that finding, because the ablation came after the ensemble was trained. (v) ES-AUC is our own implementation. (vi) We did not implement the published FIN/FIS/FIA methods or compare with published numbers.

5 Conclusions

An ensemble of eight ImageNet-pretrained two-branch EfficientNet-B0 networks, one branch per imaging modality, trained on different data splits, reaches 0.834 test AUC (95% CI 0.816–0.852) for

glaucoma detection from SLO and OCT fundus images, against 0.717 for a from-scratch ResNet18 baseline. Using both modalities is worth about 0.06–0.07 AUC over either alone, and averaging models trained on different splits adds about 0.03 over its members. Performance is similar across gender and race groups.

The strength of the approach is its simplicity: standard pretrained backbones, concatenation fusion and probability averaging. Its weaknesses are that the small test set cannot separate the ensemble from the best single model, that eight networks cost eight times as much to run, and that our best-found training recipe (flip-only augmentation) was identified too late to be included. **Future work:** combine the helpful recipe changes and test them with several seeds; implement FIN, FIS and FIA faithfully and compare on the full dataset; fine-tune retinal foundation models; treat the two image types as source and target domains with domain-adaptation methods, since transfer between them is poor; and use cross-validation over the full data to reduce noise from a single test split.

References

[1] Y. Tian, C. Wen, M. Shi, M. M. Afzal, H. Huang, M. O. Khan, Y. Luo, Y. Fang, and M. Wang. FairDomain: Achieving fairness in cross-domain medical image segmentation and classification. In *ECCV*, 2024. arXiv:2407.08813.

[2] Y. Luo et al. Harvard Glaucoma Fairness: A retinal nerve disease dataset for fairness learning and fair identity normalization. *IEEE Trans. Medical Imaging*. arXiv:2306.09264.

[3] Y. Luo, Y. Tian, M. Shi, T. Elze, and M. Wang. Harvard Eye Fairness: A large-scale 3D imaging dataset for equitable eye diseases screening and fair identity scaling. arXiv:2310.02492, 2023.

[4] Y. Luo et al. FairCLIP: Harnessing fairness in vision-language learning. In *CVPR*, 2024. arXiv:2403.19949.

[5] FairEnc: A fair vision-language model with fair vision and text encoders for glaucoma detection. arXiv:2605.04882.

[6] Fair-Eye Net: A fair, trustworthy, multimodal integrated glaucoma full chain AI system. arXiv:2601.18464.

[7] RED-Sphere: Hyperspherical residual edge debiasing for cross-population fundus disease domain generalization. arXiv:2607.10777.

[8] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *CVPR*, 2016.

[9] M. Tan and Q. V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In *ICML*, 2019.

[10] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger. Densely connected convolutional networks. In *CVPR*, 2017.

[11] I. Radosavovic, R. P. Kosaraju, R. Girshick, K. He, and P. Dollár. Designing network design spaces. In *CVPR*, 2020.

[12] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie. A ConvNet for the 2020s. In *CVPR*, 2022.

[13] A. Dosovitskiy et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021.

[14] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou. Training data-efficient image transformers and distillation through attention. In *ICML*, 2021.

[15] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo. Swin Transformer: Hierarchical vision transformer using shifted windows. In *ICCV*, 2021.

[16] Z. Dai, H. Liu, Q. V. Le, and M. Tan. CoAtNet: Marrying convolution and attention for all data sizes. In *NeurIPS*, 2021.

[17] Z. Tu, H. Talebi, H. Zhang, F. Yang, P. Milanfar, A. Bovik, and Y. Li. MaxViT: Multi-axis vision transformer. In *ECCV*, 2022.

[18] M. Oquab et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.

[19] O. Russakovsky et al. ImageNet large scale visual recognition challenge. *IJCV*, 2015.

[20] R. Wightman. PyTorch Image Models (timm). `https://github.com/huggingface/pytorch-image-models`, 2019.

[21] A. Paszke et al. PyTorch: An imperative style, high-performance deep learning library. In *NeurIPS*, 2019.

[22] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. In *ICLR*, 2019.

[23] L. N. Smith and N. Topin. Super-convergence: Very fast training of neural networks using large learning rates. In *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications*, SPIE, 2019.

[24] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz. mixup: Beyond empirical risk minimization. In *ICLR*, 2018.

[25] S. Yun, D. Han, S. J. Oh, S. Chun, J. Choe, and Y. Yoo. CutMix: Regularization strategy to train strong classifiers with localizable features. In *ICCV*, 2019.

[26] G. Huang, Y. Sun, Z. Liu, D. Sedra, and K. Q. Weinberger. Deep networks with stochastic depth. In *ECCV*, 2016.

[27] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna. Rethinking the Inception architecture for computer vision. In *CVPR*, 2016.

[28] Y. Ganin and V. Lempitsky. Unsupervised domain adaptation by backpropagation. In *ICML*, 2015.

A Overview of all experiments

Besides the final model we ran four batches of experiments (51 training runs plus ensembles) on the same data, split and validation protocol. Each batch had a question; Table 5 lists them and the following appendices give the details. All runs use the validation split described in Section 4.1 to choose the epoch; nothing was selected on test AUC automatically.

B Backbone and modality comparison (sweep 1)

Motivation. Before committing to a design we wanted to know how much pretraining, the architecture family (CNN, transformer, hybrid) and the use of both modalities matter.

Appendix	Question (motivation)	Setup	Main outcome
B Backbones	Which architecture family and which modality choice work best?	13 configurations, plain recipe, 15 epochs	Pretrained 0.747–0.827 vs scratch 0.717; EfficientNet-B0 best (0.827); both modalities help
C Training recipe	How to fight overfitting on 2,550 images?	Stronger recipe on Swin-Tiny and EfficientNet-B0; mixup, drop-path, LR, schedule; 4 ablations	Drop-path helps (0.817 vs 0.798); strong geometric augmentation hurts (flip-only 0.819)
D Fusion / foundation	Is smarter fusion or a self-supervised ViT better?	Gated fusion, cross-attention, DINOv2-S on Swin-scale recipe	No gain: 0.812, 0.795, 0.809 vs 0.808 ± 0.004
E Fairness losses	Can fairness-aware training raise worst-group performance?	Group reweighting, FIS-like loss, adversarial race head	No consistent gain (0.787, 0.806, 0.805)
F Domain shift, tabular	How different are the two modalities? Do clinical variables help?	Train on one modality, test on the other; add age / MD	Transfer 0.68–0.70; age no gain; MD 0.958 (not comparable)
G Ensembles	Does averaging help, and which members?	Eight ensemble compositions, with and without TTA	0.816–0.834

Table 5: The experiments behind and around the final model.

Setup. Thirteen configurations were trained with the *plain* recipe: cross-entropy, AdamW with a one-cycle schedule, 15 epochs, horizontal flips and mild brightness jitter, with the per-family learning rates in Table 6. Fusion was late (one backbone per modality) unless marked "early" (a 2-channel input to one backbone). Single-modality baselines use one branch only. Epochs are selected by validation AUC.

Backbone	Peak LR	Batch size	Weight decay
ResNet18 (scratch)	3e-4	64	1e-2
ResNet50	2e-4	32	1e-2
EfficientNet-B0	5e-4	32	1e-2
DenseNet121	2e-4	32	1e-2
ConvNeXt-Tiny	1e-4	32	5e-2
ViT-S / DeiT-S / Swin-T	5e-5	32	5e-2
CoAtNet-0	1e-4	32	5e-2
MaxViT-Tiny	5e-5	16	5e-2

Table 6: Hyper-parameters of the first sweep.

Results. Table 11 and Figure 5 show the results. Every pretrained model improved on the from-scratch ResNet18 (0.717): EfficientNet-B0 0.827, Swin-Tiny 0.807, DenseNet121 0.803, CoAtNet-0 0.793, ViT-Small 0.790, MaxViT-Tiny 0.778, DeiT-Small 0.768, ResNet50 0.768 (early fusion 0.747) and ConvNeXt-Tiny 0.767. The confidence intervals of the models between 0.79 and 0.83 overlap almost entirely, so no architecture family is established as best. Modality: with EfficientNet-B0, SLO alone gives 0.758, OCT alone 0.764 and both 0.827 (Figure 6). Late fusion was slightly better than early fusion for ResNet50 (0.768 vs 0.747, within the intervals). Swin-Tiny had the best *validation* AUC (0.844) but not the best test AUC, a reminder that the 450-image validation set is itself noisy; this is why the training-recipe study (Appendix C) uses seeds.

Takeaway. Pretrained, two-modality models work; the choice of family is secondary. We took EfficientNet-B0 forward because it was the best and the cheapest strong model (about 5 minutes per run).

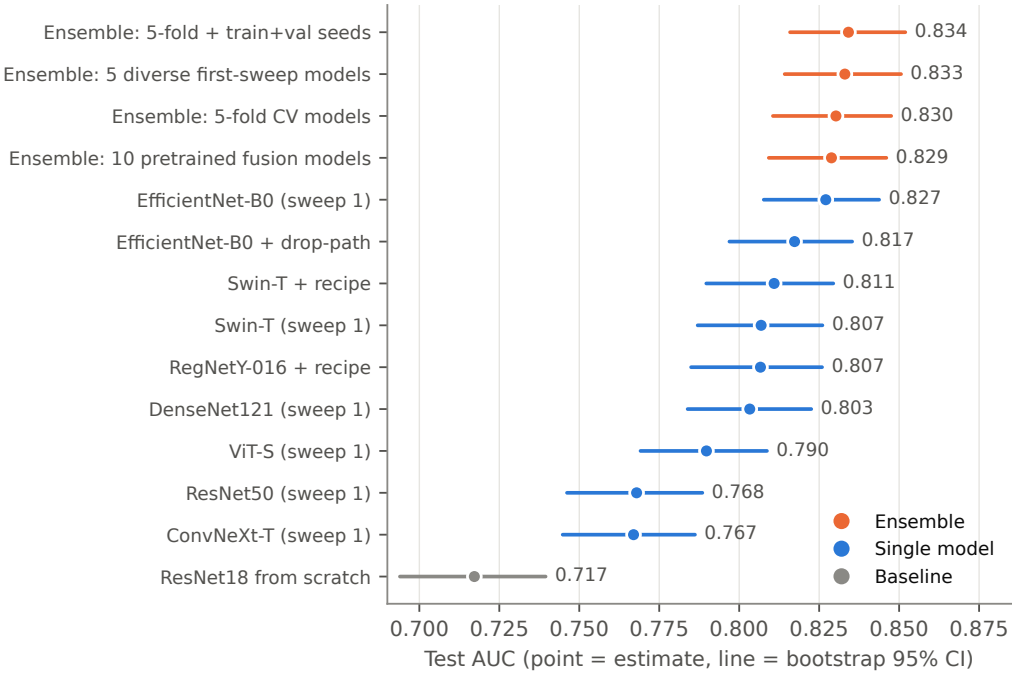


Figure 5: Test AUC with bootstrap 95% confidence intervals for the main single models and the best ensembles.

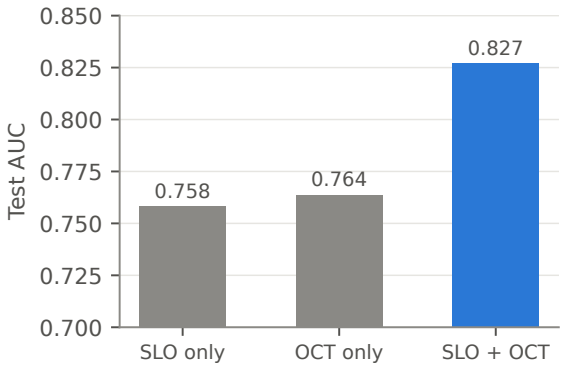


Figure 6: Contribution of each modality (EfficientNet-B0, plain recipe).

C Training recipe study and ablations (sweeps 2–3)

Motivation. All models overfit: the plain EfficientNet-B0 reaches a training loss of about 0.008 while validation AUC stops improving early (Figure 7). We therefore tried to regularise harder.

Setup. The *stronger recipe* adds on-GPU geometric augmentation (rotation ±15°, scale ±10%, shift ≤ 5%), label smoothing 0.1, an EMA of the weights, a 5× learning rate on the new head, and 25 epochs. It was first applied to Swin-Tiny (the backbone with the best validation AUC in sweep 1), with three seeds, and then to EfficientNet-B0 with variants: a 12-epoch schedule, mixup/CutMix, drop-path 0.2 with head dropout 0.5, a lower learning rate (2×10^{-4}), and gated fusion; plus four neighbouring backbones (EfficientNet-B2/B3, RegNetY-016, DenseNet121). The configuration for the final model was chosen by a fixed rule on validation AUC: a challenger replaces the plain stronger recipe only if its validation AUC exceeds the three-seed mean by more than max(0.005, seed s.d.). Drop-path won (0.852 against 0.832 mean, margin 0.013).

Results. Table 12 (full list) and Table 7 summarise this. The stronger recipe did not help uniformly: Swin-Tiny was unchanged (0.808 ± 0.004 over three seeds, against 0.807 in sweep 1), but EfficientNet-B0 *dropped* from 0.827 to 0.798 ± 0.003. Drop-path with dropout helped (0.817), while mixup/CutMix (0.800), a lower learning rate (0.790) and a shorter schedule (0.777) did not; the other backbones were no better than EfficientNet-B0 (B2 0.796, B3 0.779, RegNetY-016 0.807, DenseNet121 0.781).

Ablation: why did the recipe hurt EfficientNet? We first suspected our EMA implementation, which copies batch-norm statistics from the live model instead of averaging them (harmful for batch-norm CNNs but not for the layer-norm Swin). We ran four single-factor ablations (one seed each; Table 7). Switching EMA *off* lowered AUC (0.786), so the hypothesis was *refuted*: EMA helps. The main culprit was the strong geometric augmentation: flips and mild jitter alone raised AUC from 0.798 to 0.819 (about 7 seed standard deviations), and a head learning rate of 1× gave a smaller gain (0.807). A plausible but untested reason is that the images are spatially registered, so rotations and shifts may destroy position-dependent cues. Restoring all three choices (our approximate sweep-1 setup) gives 0.811, below the original 0.827, so part of that figure is probably single-seed luck. The ablations vary one factor each with a single seed, so the ordering of their effects is indicative. We did not run the combination of the helpful changes (flip-only augmentation, drop-path, head learning rate 1×); this is the most promising next experiment.

Takeaway. Regularisation matters but most standard tools did nothing; drop-path was the one clear win among those tried, and geometric augmentation was harmful.

EfficientNet-B0 variant	Val AUC	Test AUC	TTA AUC
Recipe (3 seeds, mean ± s.d.)	0.832	0.798 ± 0.003	0.803
EMA off	0.817	0.786	0.792
Head LR ×1 (instead of ×5)	0.828	0.807	0.807
Flip + mild jitter only (no rotate/scale/shift)	0.853	0.819	0.824
All three reverted, no label smoothing, 15 epochs (approx. sweep-1 setup)	0.844	0.811	0.814
Sweep-1 run (original code, one seed)	0.833	0.827	0.828

Table 7: Single-factor ablations of the stronger recipe (EfficientNet-B0, one seed each; the recipe row is the three-seed mean). The last row is the original first-sweep run.

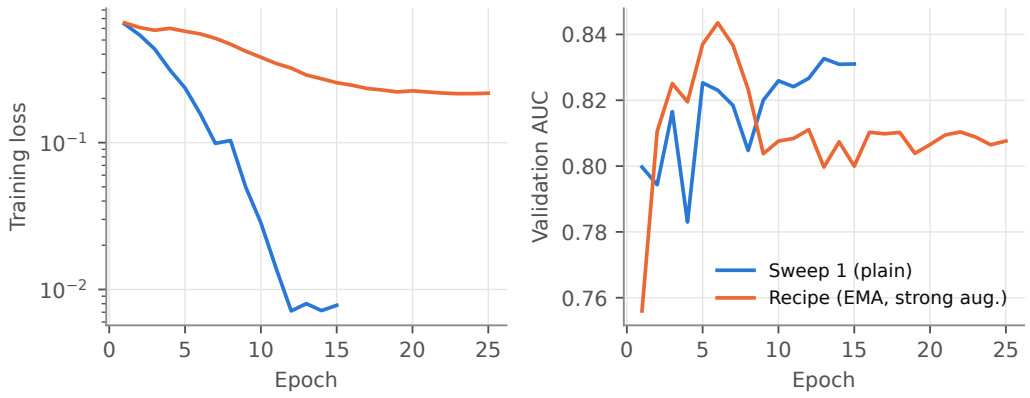


Figure 7: EfficientNet-B0: training loss (log scale) and validation AUC. The stronger recipe's loss stays higher because of label smoothing and augmentation, but its validation AUC peaks around epoch 6 and then declines.

D Fusion variants and a foundation model (sweep 2)

Motivation. Plain concatenation may not be the best way to combine two modalities, and a large self-supervised model might transfer better than ImageNet weights.

Setup. On the Swin-Tiny recipe (25 epochs): *gated fusion* with modality dropout, $\mathbf{z} = \sigma(W[\mathbf{p}_S; \mathbf{p}_O]) \odot \mathbf{p}_S + (1 - \sigma(\cdot)) \odot \mathbf{p}_O$ with $\mathbf{p} = W'\mathbf{z}$ (one modality's feature zeroed for 20% of training samples); *cross-attention fusion* with ViT-Small (each modality's tokens attend to the other's before the class tokens are concatenated); and a fine-tuned *DINOv2-Small* [18] with late fusion. Figure 8 sketches the variants.

Results. (Table 9.) Gated fusion 0.812, cross-attention 0.795 and DINOv2 0.809, against 0.808 ± 0.004 for plain concatenation: no variant is distinguishable from the baseline. A gated variant on EfficientNet-B0 scored 0.810 against 0.798.

Takeaway. Plain concatenation is sufficient at this data scale, so the final model keeps it.

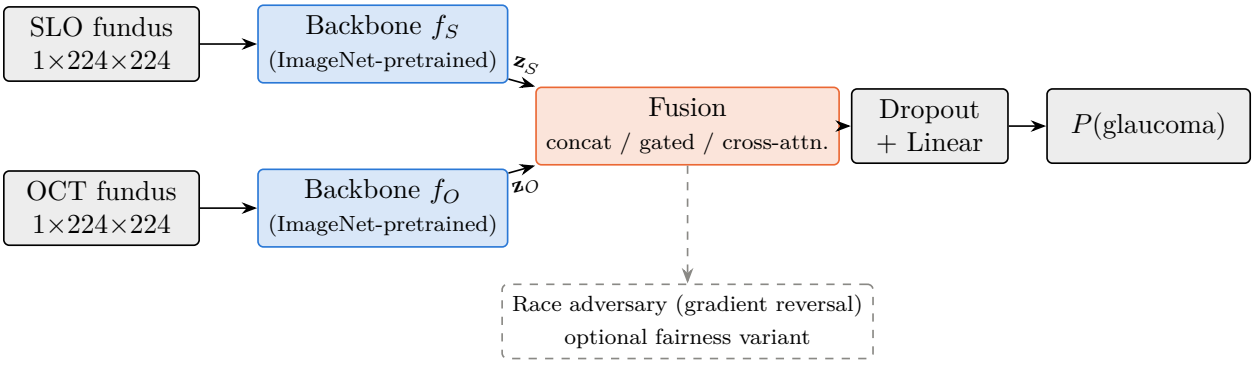


Figure 8: Variants evaluated in the appendices: the fusion block is replaced by gated or cross-attention fusion; a single backbone with a 2-channel input (early fusion) or with one modality only; an adversarial race head on the fused feature.

E Fairness-aware training and per-group results (sweep 2)

Motivation. The data are demographically imbalanced, and fairness-aware losses are a central theme of the dataset literature [2, 3, 1].

Setup. On the Swin-Tiny recipe: (i) *group reweighting*: each sample is weighted by the inverse frequency of its (race, label) cell; (ii) a *FIS-like loss*: per-sample weights equal to the product of the individual loss (relative to the batch mean) and its race group's mean loss (relative to the batch mean), clipped to $[0.2, 5]$ and renormalised, which is our approximation of the idea behind Fair Identity Scaling [3] and not an exact reimplementation; (iii) *adversarial debiasing*: a small classifier predicts race from the fused feature through a gradient-reversal layer [28] (loss weight 0.5). The adversarial variant was repeated on EfficientNet-B0.

Results. (Table 9.) The FIS-like loss (0.806) and adversarial debiasing (0.805) were within noise of the plain recipe (0.808); group reweighting was lower (0.787). On EfficientNet-B0 the adversarial variant scored 0.802 against 0.817 for the configuration it extends. Table 8 shows within-group AUC for representative models and ensembles. Differences between gender and race groups are about 0.01–0.03 for most models; the Hispanic group (83 test images) shows the largest and least stable differences (0.74–0.81), which sampling noise could explain.

Takeaway. No fairness-specific loss gave a reliable benefit; the best group-level behaviour came from ensembling.

n (test)	Female 1449	Male 551	Asian 177	Black 301	White 1522	Non-Hisp. 1917	Hispanic 83
EfficientNet-B0 (sweep 1)	0.824	0.835	0.823	0.816	0.827	0.830	0.740
Swin-T + recipe (sweep 2)	0.813	0.806	0.832	0.811	0.806	0.812	0.782
EfficientNet-B0 + drop-path (sweep 3)	0.816	0.822	0.831	0.803	0.814	0.819	0.811
5-model ensemble (sweep 1)	0.833	0.835	0.829	0.826	0.830	0.834	0.796
8-model ensemble (sweep 3)	0.832	0.841	0.856	0.821	0.829	0.836	0.806

Table 8: Test AUC within demographic groups for representative models and ensembles.

F Domain shift and tabular inputs (sweep 2)

Motivation. FairDomain treats image types as domains [1]. If SLO and OCT carry different information, a model trained on one should transfer poorly to the other, which would explain why fusion helps. Separately, we asked whether clinical variables add anything.

Setup. *Cross-domain:* a single Swin-Tiny branch is trained on one modality only and evaluated on the same modality and on the other (fed through the same single-channel input). *Tabular:* age (and, as an upper bound, the visual-field mean deviation MD) is passed through a small MLP and concatenated to the fused feature.

Results. (Table 9.) Trained on SLO, the model reaches 0.794 on SLO but 0.677 on OCT; trained on OCT, 0.786 on OCT and 0.701 on SLO. Adding age gave 0.808, no change. Adding MD raises AUC to 0.958, because MD is a clinical measure closely tied to the label; it is not a fair comparison and is excluded from every other comparison.

Takeaway. The two modalities are complementary (and domain adaptation between them is an open problem); age adds nothing.

Variant (Swin-Tiny backbone)	Val AUC	Test AUC	95% CI	ES-AUC race	Worst-group	Other-domain AUC
+ age and MD (upper bound)	0.963	0.958	[0.950, 0.965]	0.938	0.941	–
Gated fusion + modality dropout	0.842	0.812	[0.793, 0.832]	0.800	0.803	–
Swin-T + recipe	0.834	0.811	[0.790, 0.831]	0.790	0.806	–
Swin-T + recipe (seed 2)	0.836	0.810	[0.789, 0.830]	0.783	0.793	–
DINOv2-S fine-tune	0.837	0.809	[0.788, 0.829]	0.760	0.786	–
+ age as input	0.840	0.808	[0.786, 0.828]	0.785	0.802	–
FIS-like loss	0.847	0.806	[0.784, 0.826]	0.779	0.800	–
Adversarial race debiasing	0.850	0.805	[0.782, 0.823]	0.788	0.801	–
Swin-T + recipe (seed 1)	0.843	0.803	[0.782, 0.823]	0.783	0.797	–
ViT-S cross-attention fusion	0.844	0.795	[0.773, 0.815]	0.772	0.786	–
Train SLO → test OCT	0.796	0.794	[0.773, 0.812]	0.759	0.786	0.677
Group-reweighted loss	0.819	0.787	[0.764, 0.808]	0.760	0.780	–
Train OCT → test SLO	0.822	0.786	[0.765, 0.804]	0.758	0.779	0.701

Table 9: Sweep 2: stronger recipe on Swin-Tiny, 25 epochs. The "+ age and MD" row is an upper bound and not a fair comparison.

G Ensembles and test-time augmentation

Motivation. With strong overfitting and noisy single runs, averaging models is the natural way to reduce variance.

Setup. We averaged probabilities of (a) the sweep-1 checkpoints (five diverse models, ten pre-trained fusion models, the top three by validation AUC, and a greedy selection on validation AUC), and (b) EfficientNet-B0 models of the final configuration: the five cross-validation members, three seeds trained on all 3,000 images, both together (the final model), and the top three configurations of the recipe study. Members were chosen in advance or by validation AUC. Each is also reported with flip TTA.

Results. (Table 10.) The ensembles range from 0.816 to 0.834 test AUC; the five diverse sweep-1 models and the final eight-model ensemble are tied (0.833 and 0.834) within the intervals. Flip TTA adds 0.001–0.010. The three full-data members alone scored 0.791–0.794 individually, no better than single models trained on the 2,550-image split, so training on more data did not help single models; the full-data models helped the ensemble slightly (0.830 to 0.834).

Takeaway. Ensembling is the most reliable gain (about +0.03 over the members); the exact composition matters little. We chose the eight-member ensemble because it uses a single, validation-selected configuration; being the highest of the compositions tried, its figure is mildly optimistic.

Ensemble (probability average)	Source	n	Test AUC	95% CI	TTA AUC	ES-AUC race
5-fold + train+val seeds (8 models)	Sweep 3	8	0.834	[0.816, 0.852]	0.835	0.802
5 diverse first-sweep models	Sweep 1 checkpoints	5	0.833	[0.814, 0.851]	0.834	0.821
5-fold CV models (drop-path config)	Sweep 3	5	0.830	[0.811, 0.848]	0.831	0.807
greedy selection on validation (8 slots)	Sweep 1 checkpoints	8	0.830	[0.811, 0.847]	0.831	0.814
10 pretrained fusion models	Sweep 1 checkpoints	10	0.829	[0.809, 0.846]	0.829	0.821
top-3 of sweep-3 tiers 1–2 by validation	Sweep 3	3	0.823	[0.803, 0.841]	0.824	0.799
top-3 by validation AUC	Sweep 1 checkpoints	3	0.819	[0.800, 0.837]	0.822	0.805
3 seeds trained on train+val	Sweep 3	3	0.816	[0.797, 0.834]	0.821	0.764

Table 10: Probability-averaging ensembles. Members were chosen in advance or by validation AUC.

H Full result tables

Model	Type	Params (M)	Val AUC	Test AUC	TTA AUC	Acc.	Worst-group AUC
EfficientNet-B0	CNN	8.0	0.833	0.827	0.828	0.763	0.816
Swin-Tiny	Transformer	55.0	0.844	0.807	0.811	0.739	0.796
DenseNet121	CNN	13.9	0.837	0.803	0.810	0.740	0.794
CoAtNet-0	Hybrid	53.3	0.835	0.793	0.796	0.716	0.787
ViT-Small	Transformer	42.9	0.841	0.790	0.792	0.724	0.781
MaxViT-Tiny (early fusion)	Hybrid	28.6	0.820	0.778	0.788	0.715	0.768
DeiT-Small (early fusion)	Transformer	21.6	0.812	0.768	0.774	0.706	0.756
ResNet50	CNN	47.0	0.796	0.768	0.779	0.703	0.741
ConvNeXt-Tiny	CNN	55.6	0.822	0.767	0.767	0.680	0.757
EfficientNet-B0, OCT only	Single modality	4.0	0.809	0.764	0.771	0.683	0.753
EfficientNet-B0, SLO only	Single modality	4.0	0.806	0.758	0.762	0.673	0.723
ResNet50 (early fusion)	CNN	23.5	0.746	0.747	0.759	0.688	0.745
ResNet18 (scratch)	CNN	22.3	0.773	0.717	0.725	0.666	0.706

Table 11: Sweep 1: 13 configurations, plain recipe, 15 epochs, validation-based epoch selection. Params = total parameters of both branches. "TTA AUC" = with horizontal-flip test-time averaging.

Compute. Single runs took from about 3 minutes (ResNet18) to 18 minutes (ConvNeXt-Tiny, 15 epochs) in sweep 1 and about 11–22 minutes in the later sweeps on the P100; EfficientNet-B0 was the cheapest of the strong models.

Configuration	Val AUC	Test AUC	95% CI	TTA AUC	ES-AUC race	Worst-group
EfficientNet-B0 + recipe	0.844	0.799	[0.779, 0.820]	0.802	0.774	0.782
seed 1	0.818	0.795	[0.774, 0.815]	0.801	0.747	0.782
seed 2	0.835	0.800	[0.778, 0.821]	0.805	0.755	0.777
12 epochs	0.836	0.777	[0.757, 0.798]	0.787	0.760	0.766
+ mixup/CutMix	0.839	0.800	[0.780, 0.819]	0.805	0.765	0.786
+ drop-path 0.2, dropout 0.5	0.852	0.817	[0.797, 0.839]	0.820	0.793	0.803
lower LR (2e-4)	0.833	0.790	[0.767, 0.812]	0.803	0.779	0.783
gated fusion + modality dropout	0.832	0.810	[0.791, 0.830]	0.814	0.801	0.803
EfficientNet-B2	0.839	0.796	[0.778, 0.817]	0.806	0.765	0.783
EfficientNet-B3	0.821	0.779	[0.759, 0.799]	0.789	0.738	0.757
RegNetY-016	0.831	0.807	[0.785, 0.828]	0.812	0.773	0.803
DenseNet121	0.828	0.781	[0.760, 0.801]	0.791	0.739	0.768
Drop-path config + adversarial race debiasing	0.847	0.802	[0.781, 0.821]	0.807	0.782	0.796

Table 12: Sweep 3: EfficientNet-B0 regularisation and backbone neighbours, 25 epochs unless stated. The drop-path configuration had the best validation AUC (0.852) and was used for the cross-validation and full-data members of the final ensemble.